Cell

II. Cell Structure & Function

Cell Theory

All living things are made of one or more cells.

Cells are the basic units of Structure & Function.

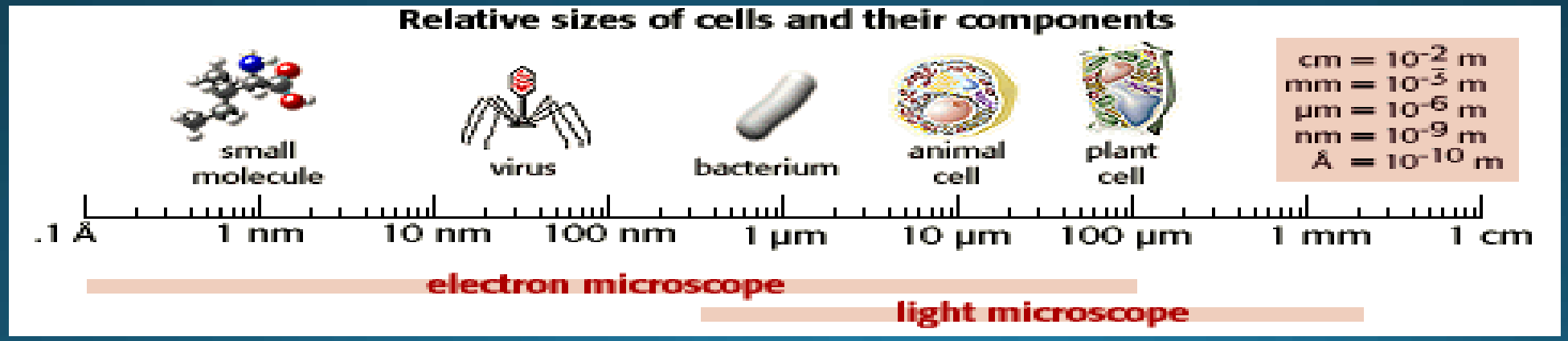
Cells originate from existing cells.

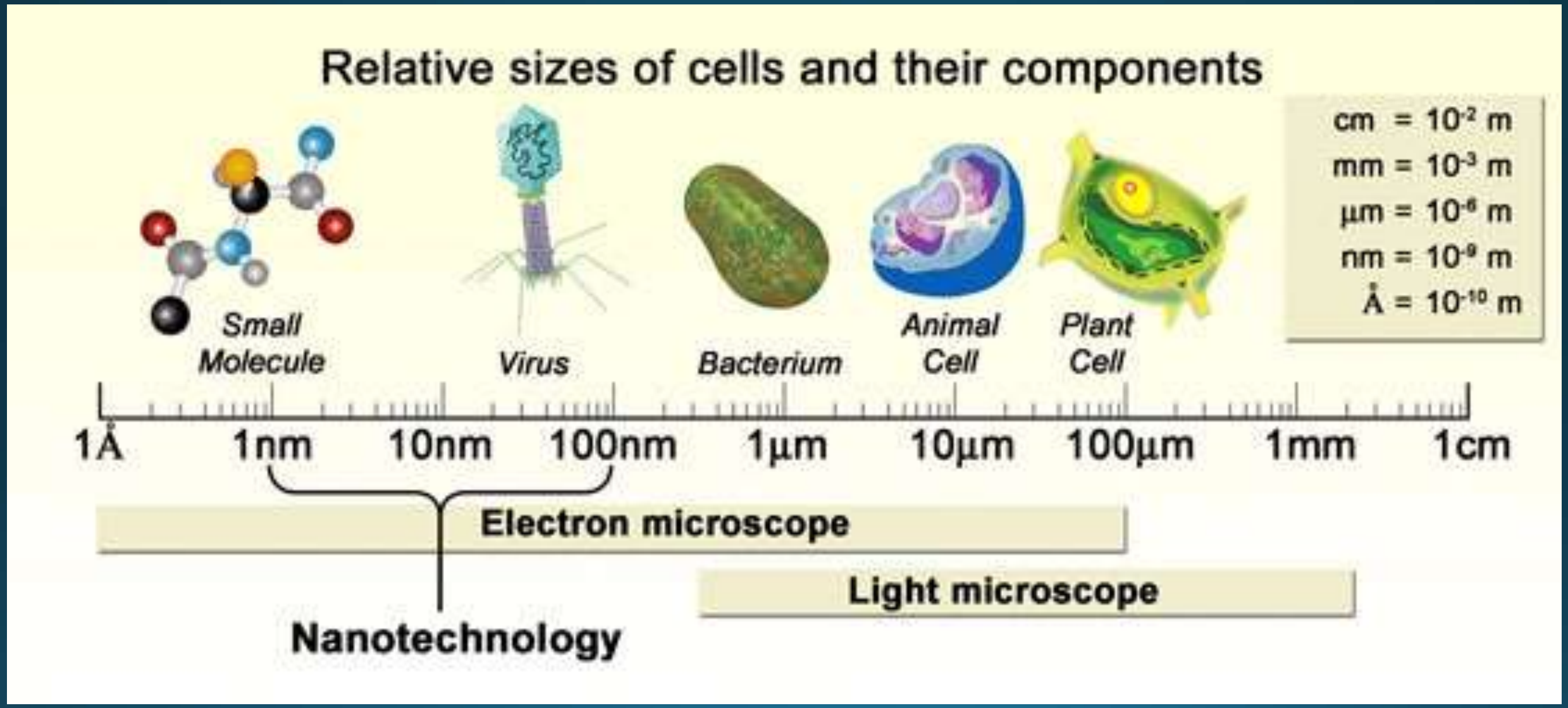
Cell Diversity

1. Not all cells are alike.

2. Cells differ in size, shape and function.

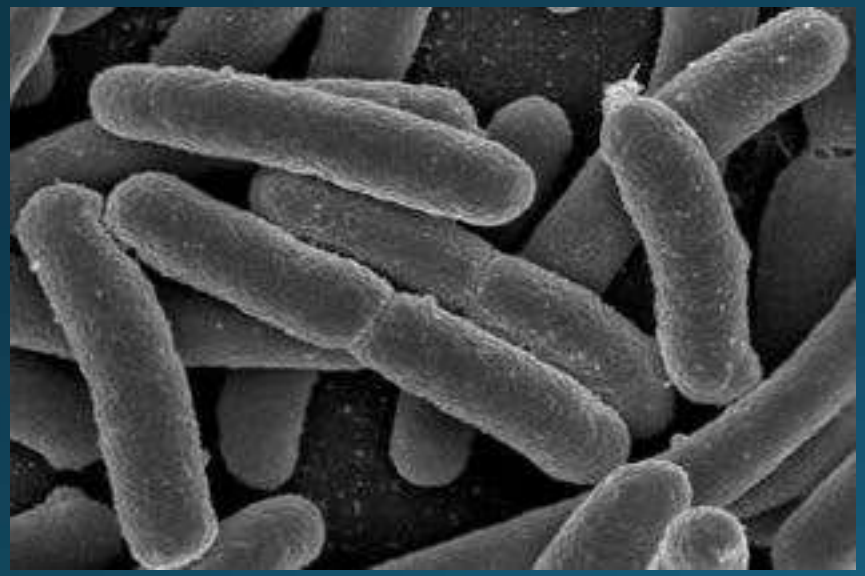
3. The female egg is the largest cell in the body & can be seen without microscope.

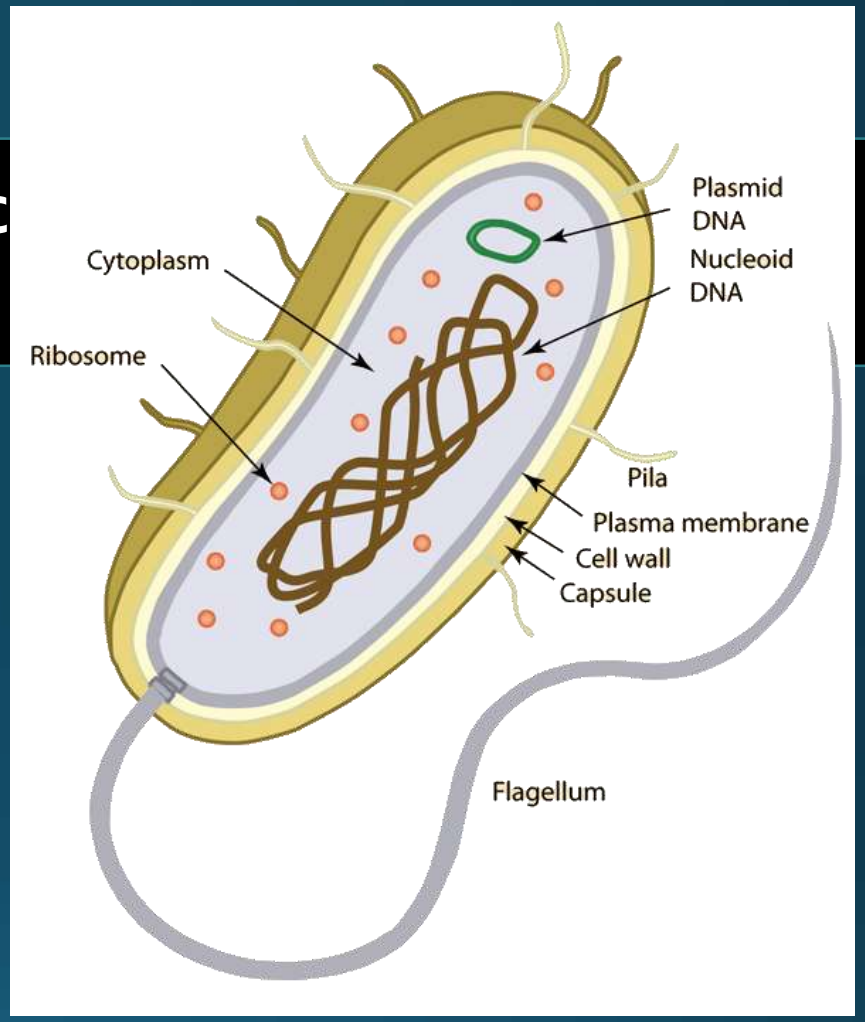



Relative sizes of cells and their components
cm = 10-2 m
mm = 10-3 m
µm = 10-6 m
nm = 10-9 m
Å = 10-10 m
Small Molecule
Virus
Bacterium
Animal Cell
Plant Cell
1Å
1nm
10nm
100nm
1µm
10µm
100µm
1mm
1cm
Electron microscope
Light microscope
Nanotechnology

Cell Diversity

4. Bacterial cells are some of the smallest c
microscope.

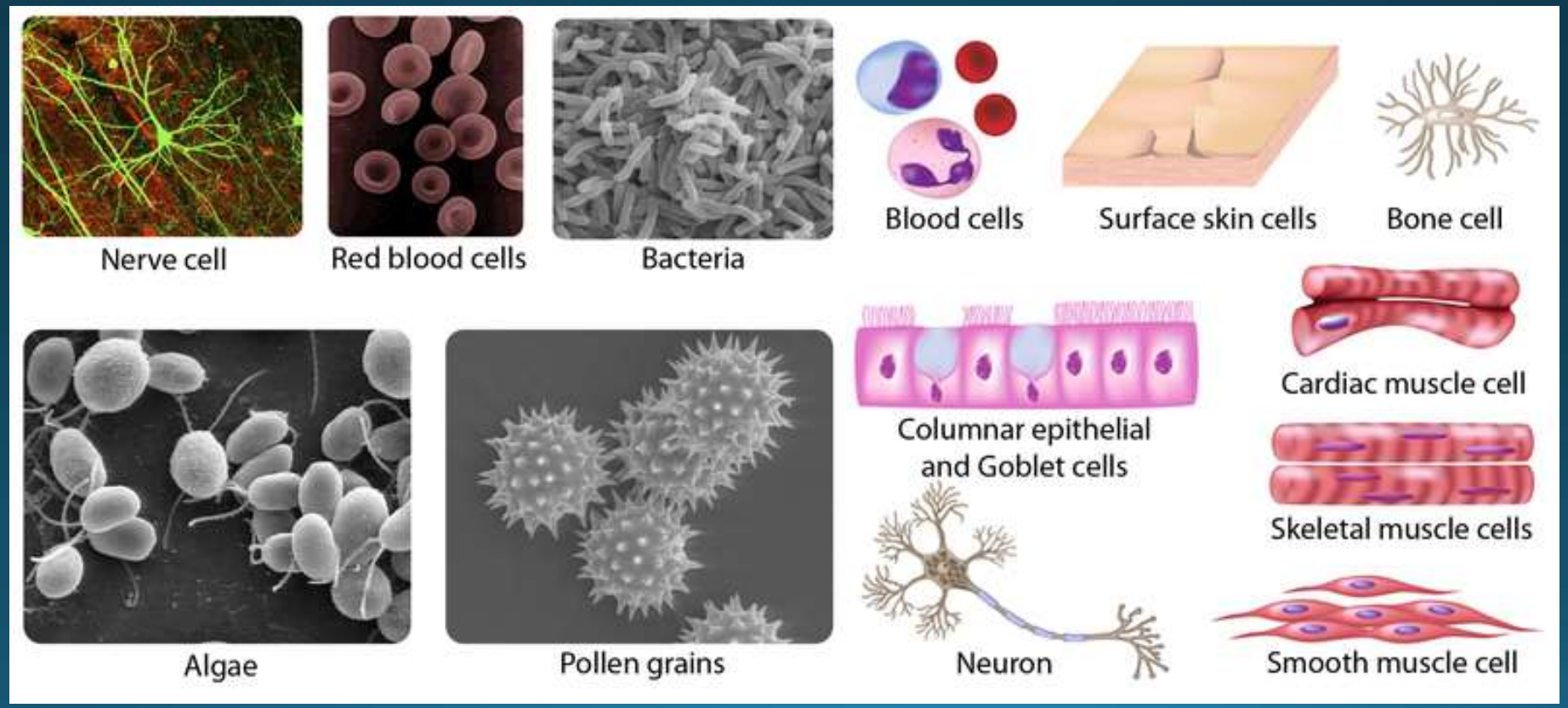




Cell Diversity

9. Cell size is also limited by the amount of cytoplasmic activity that the cell's nucleus control.

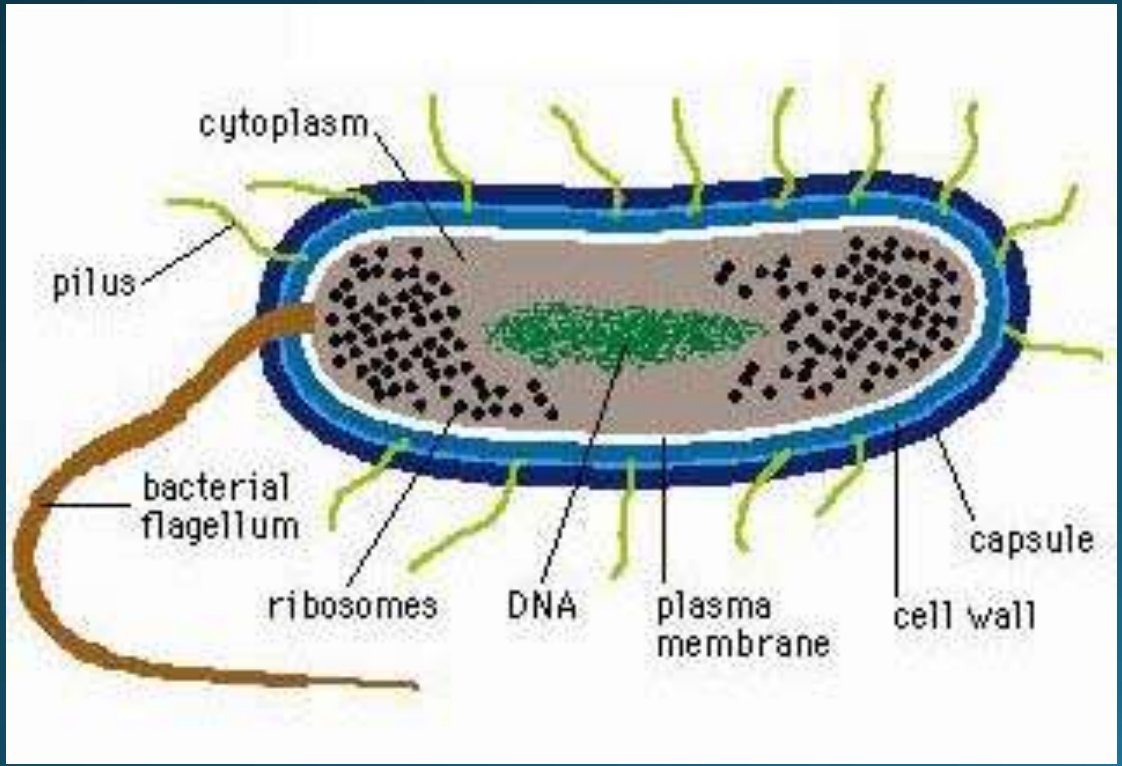
10. Cells come in a variety of shapes & the shape helps determine the function of the cell (e.g. Nerve cells and red blood cells).



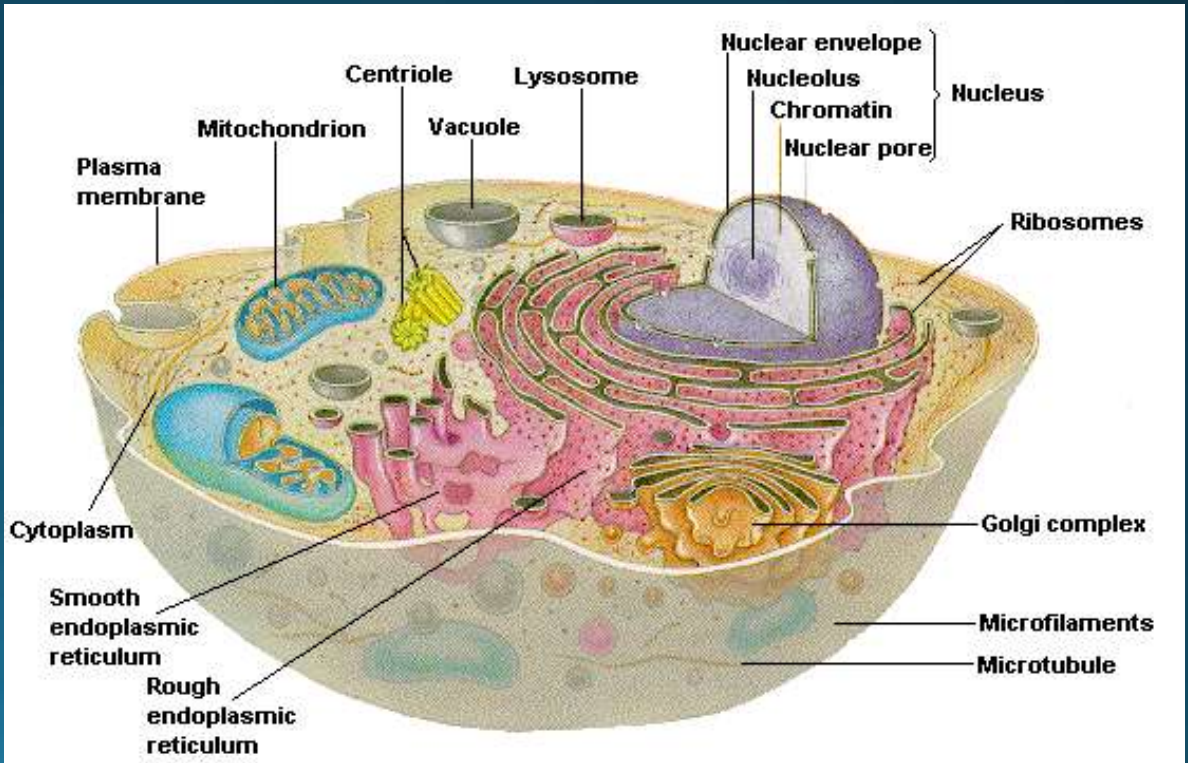
Cell diversity

Cell Types

Prokaryotes



Eukaryotes



1. Prokaryotes

Prokaryotic cells are less complex

Unicellular.

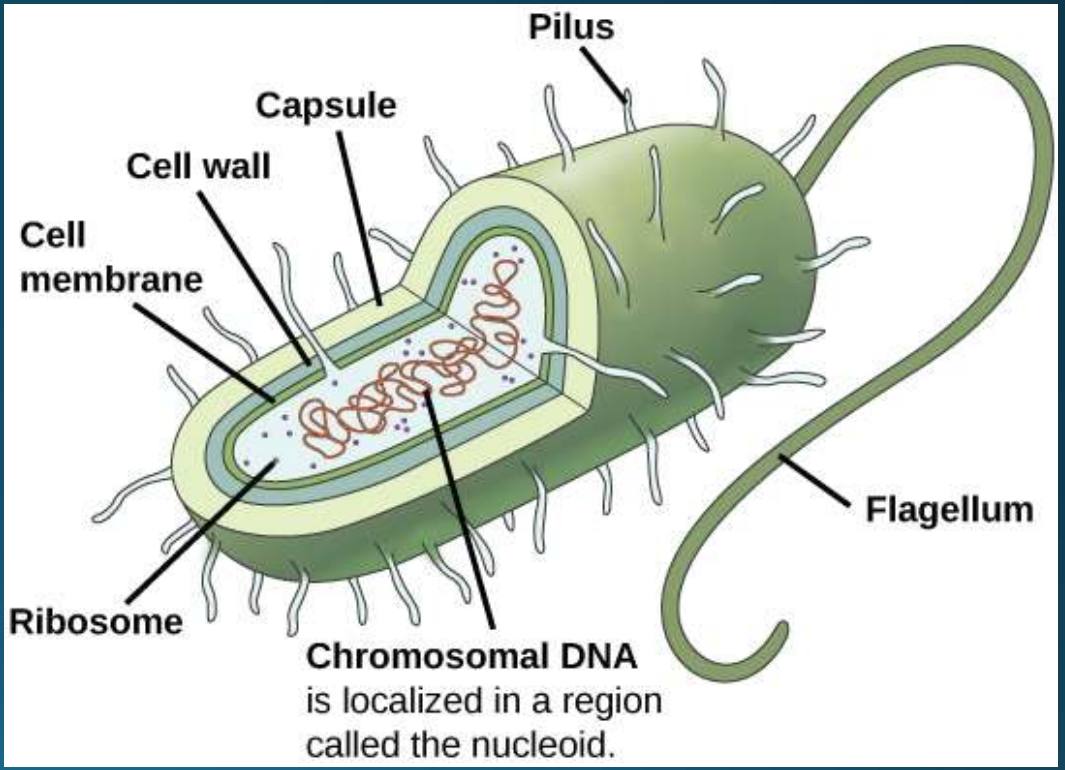
Don’t have a nucleus & No membrane bound organelles.

Most have a cell wall surrounding the cell membrane& a single looped chromosome.

Include bacteria & blue green bacteria.

Found in kingdom Monera.

Prokaryotes or bacteria lack a nucleus.



2. Eukaryotes

More complex cells.

Include both unicellular & Multicellular.

Do have a true nucleus and membrane bound organelles.

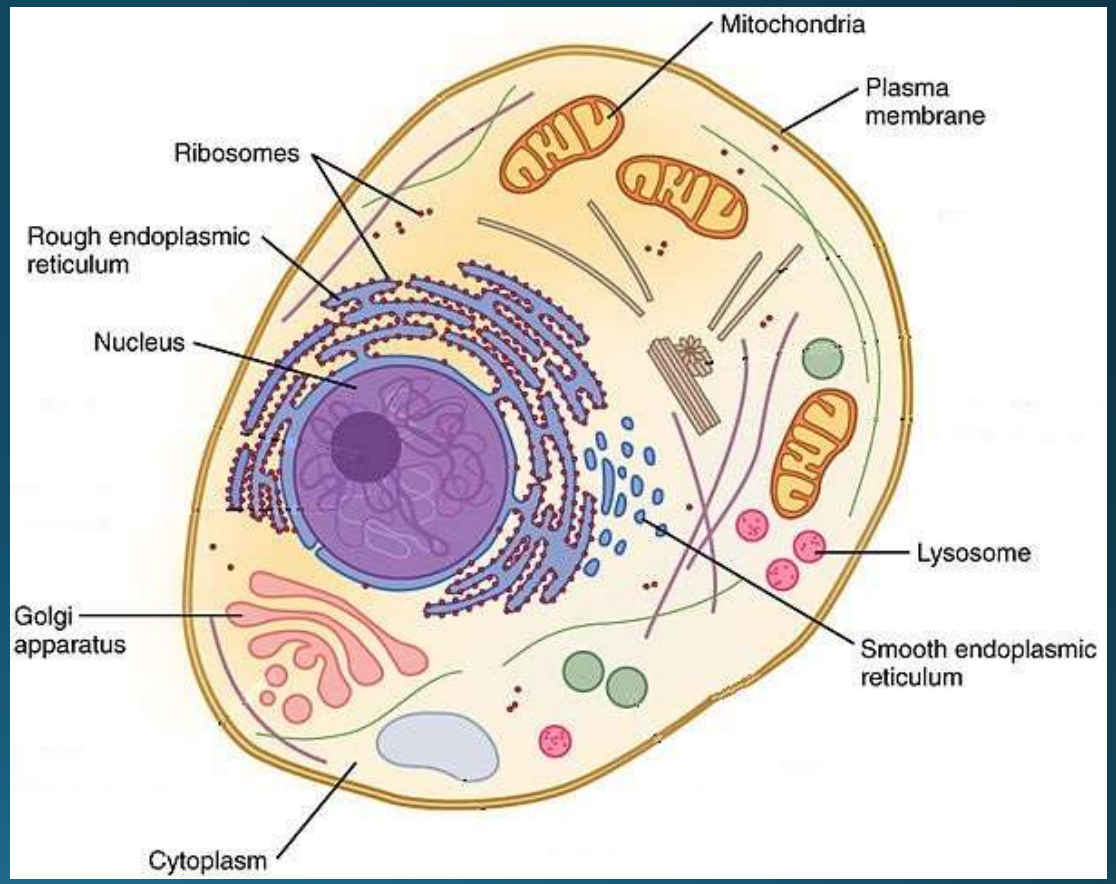
Organelles are internal structures that perform specific function.

Organelles Surrounded by single or double membrane.

Eukaryotic cells surrounded by a thin cell membrane that controls what leaves or enters the cell.

Nucleus located in the center of the cell, contains genetic material & controls activity of the cell.

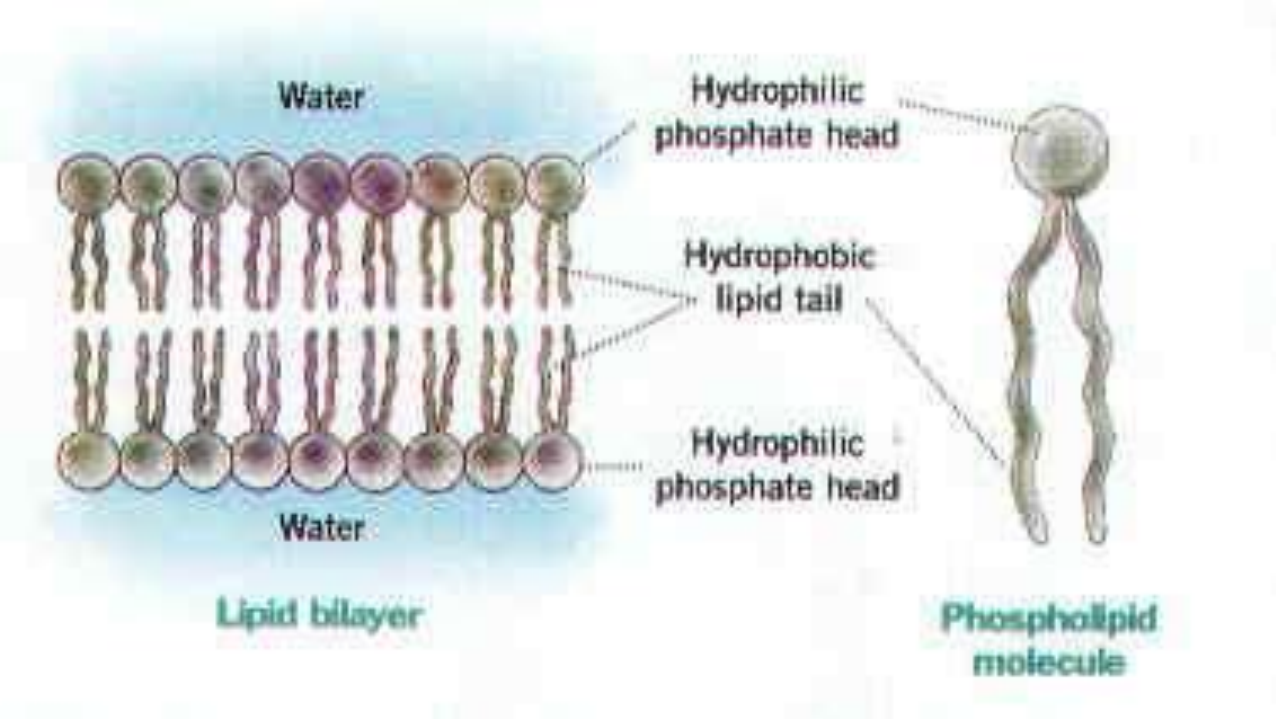
Eukaryotes includes plant cell, animal cell, fungi, algae & Protests.



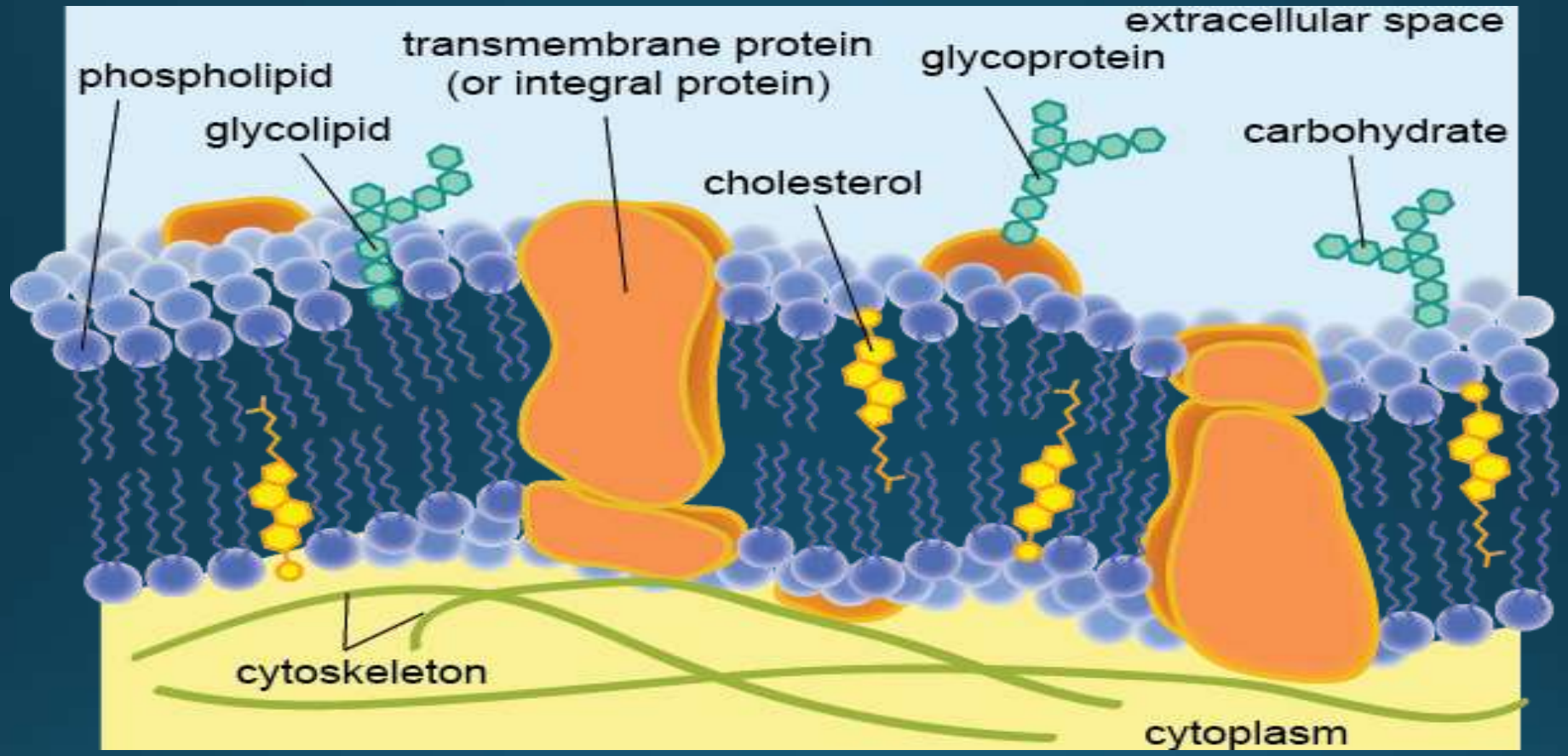
Structure & Function of Cell Organelles

I. Cell Membrane

It composed mainly from a lipid bi-layer made of phospholipids molecules in addition to protein and carbohydrate content



Fluid Mosaic Model



Phospholipid & Proteins can move side to side making the membrane appear fluid.

Proteins embedded in cell membrane form a pattern or mosaic.

Functions of the cell membrane

Separates the cytoplasm from external environment and gave the cell its characteristic shape.

Protects the cell & Regulate exchange of materials with surroundings (Cell transport).

Cell membranes are selectively permeable only allowing certain materials to enter/leave.

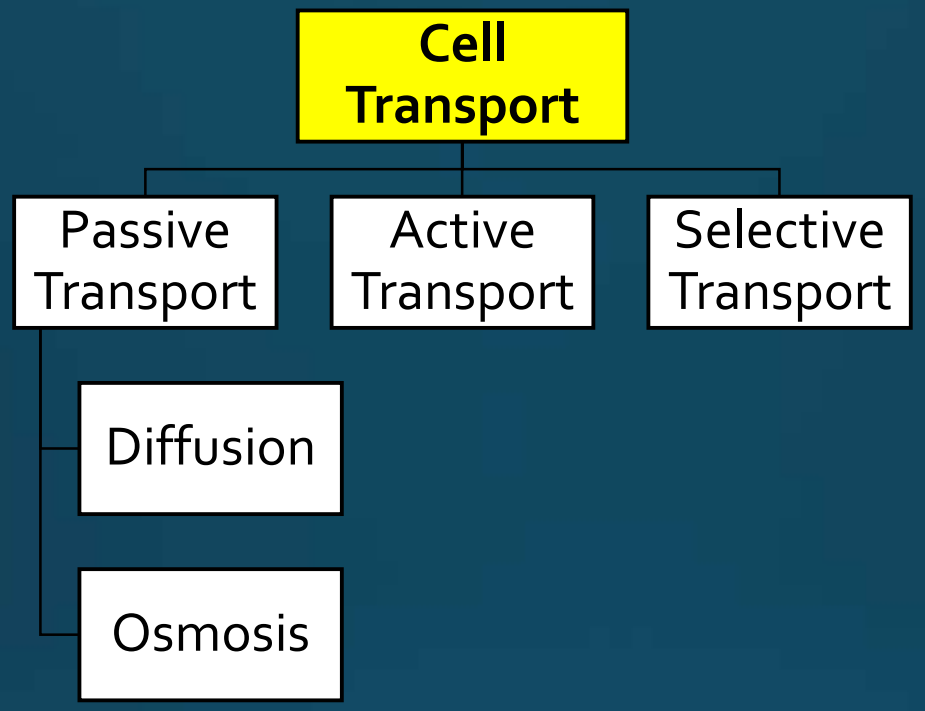
Cell membrane proteins help moving materials in/out the cells (Ion Protein Channels, Transmembrane proteins binding substance, Carbohydrate binding chemical signals) .

Endocytosis and Exocytosis.

Sodium - Potassium pump which is responsible for polarization of cell membrane.

Cell membrane take part in formation (Myline sheath, Microvilli, Cilia and Flagella)

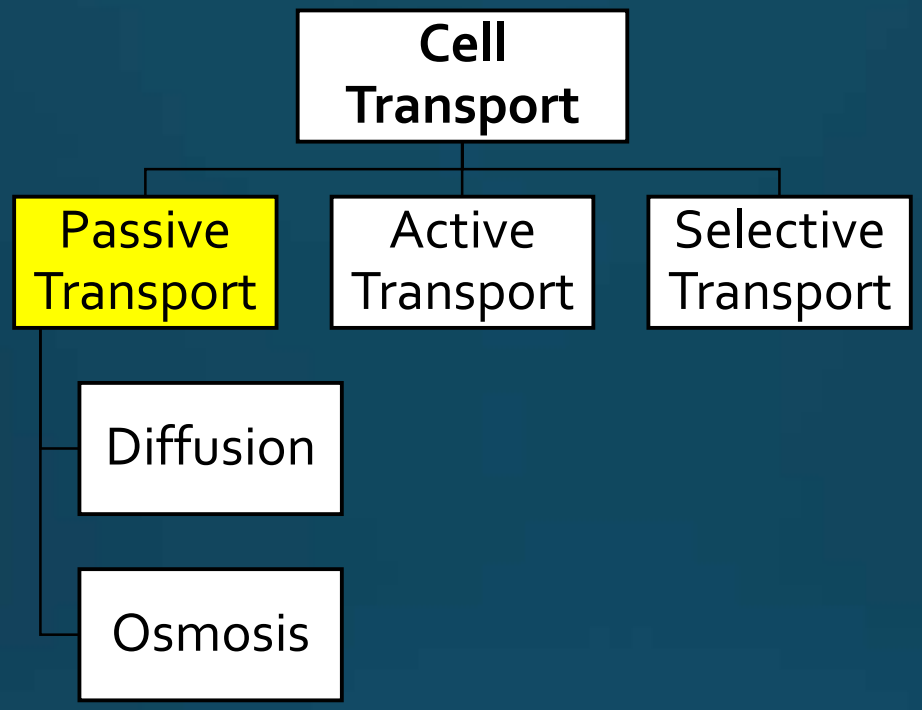
Cell Membrane



Cell Transport

Molecules tend to move from higher concentration to lower concentrations until it reach homeostasis.

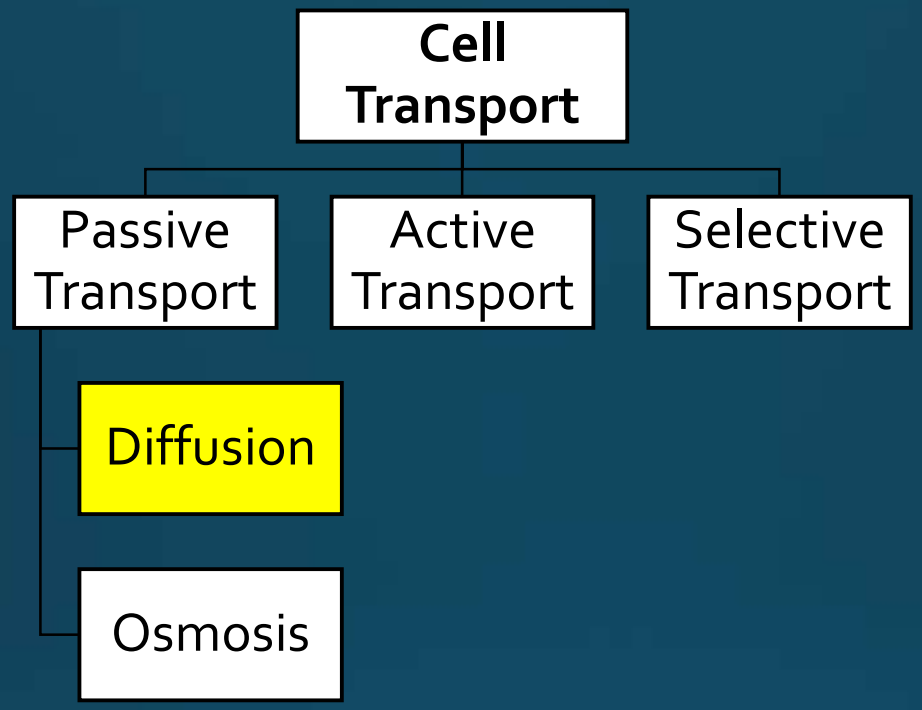
The cell membrane is selectively preamble so it controls the movement of substances in/out.



Passive Transport

In the movement of molecules from higher concentration to lower concentrations without use energy.

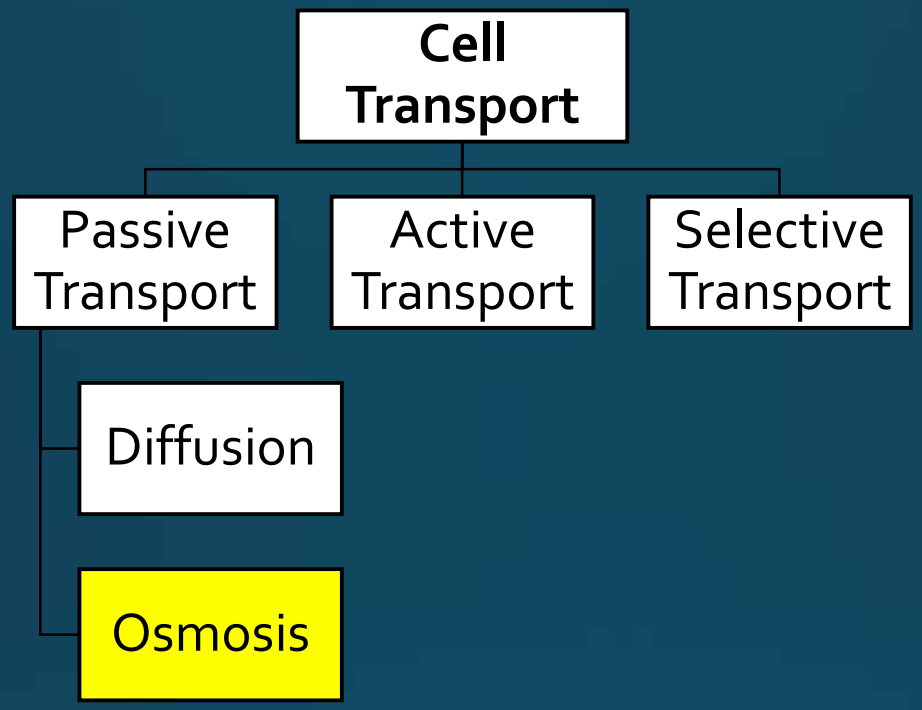
Movement is due to unequal concentration inside and outside the cell.



Diffusion

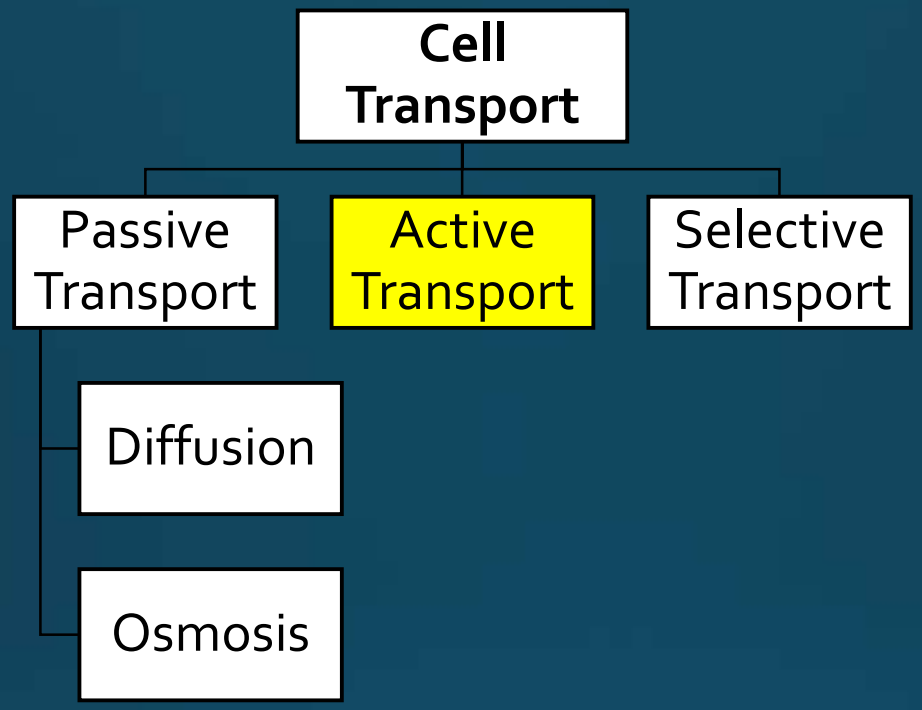
Diffusion is one principle method of movement of small molecules within cells as Co_2, Oxygen.

Gas exchange in gills of fishes and lungs of animal, is operated by diffusion.

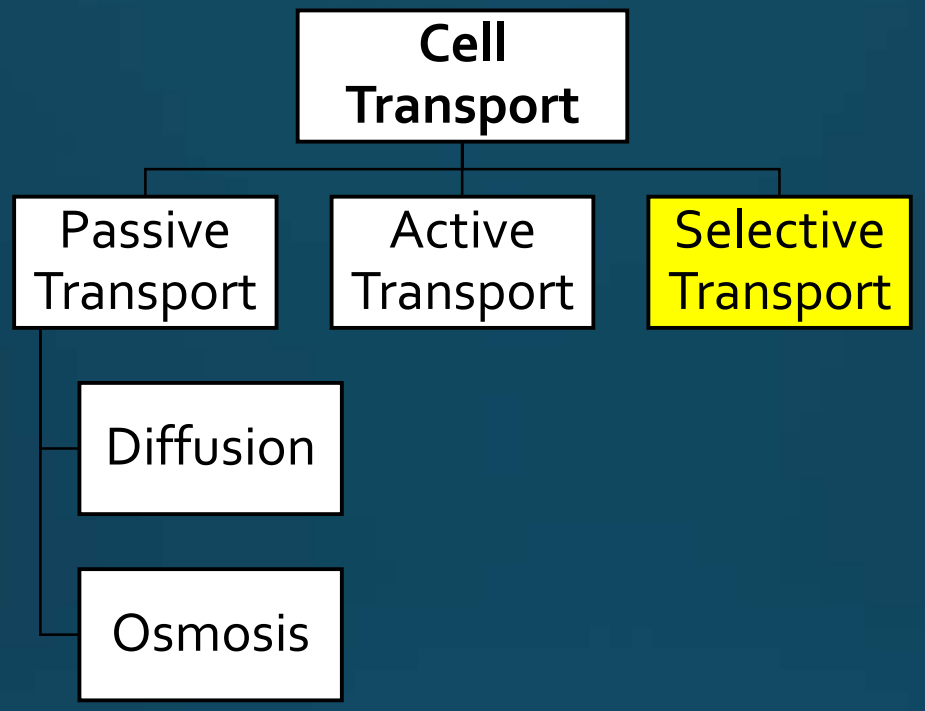


Osmosis

Is the diffusion of water a cross a semi-preamble, preamble and selective preamble membrane.



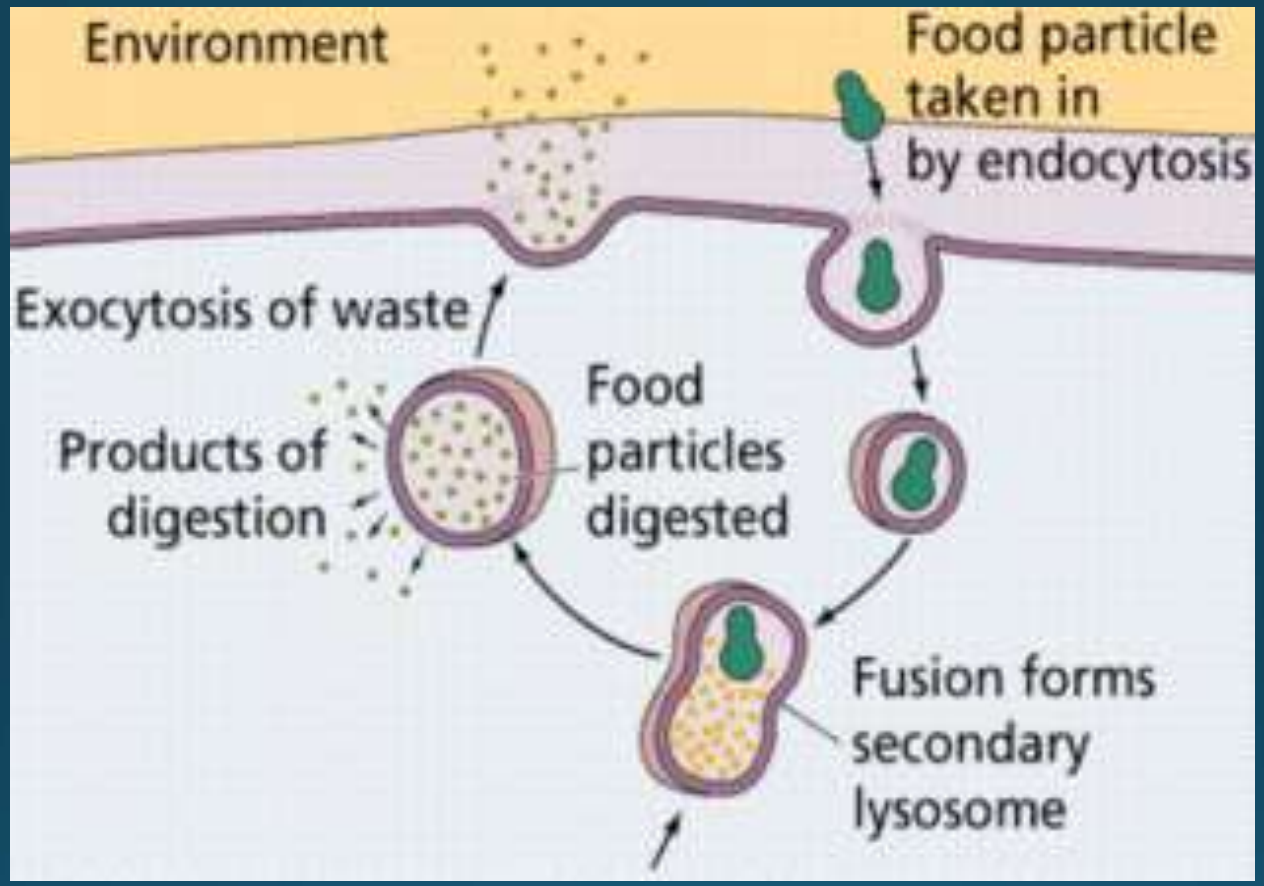
Active Transport	Large molecules (Sugar/Amino acids) are carried into/out with use of energy. Also during transportation from lower to higher concentrations. In this case, Proteins have to move against the concentration gradient (Sodium-potassium pump in nerve cells).



Selective Transport

The presence of cell receptors allows the cell to select and to determine which material enter the cell and which to be shut out.

Endocytosis & Exocytosis



Taking solid matters in the surroundings into the cytoplasm called Endocytosis.

Taking fluid droplets in from the surroundings into the cytoplasm called Pinocytosis.

Exocytosis is reversed to Endocytosis (extrude membrane vesicle from cytoplasm).